

Algebra II Conic Sections Cheatsheet

Module 11

Identification

For non-rotated conics:

$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

circle $A = C$

parabola one squared variable

ellipse x^2, y^2 same sign, unequal coefficients

hyperbola x^2, y^2 opposite signs

Completing the Square

$$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$$

Group x terms and y terms separately.

Circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Center:

$$(h, k)$$

Radius:

$$r$$

Parabola

Vertical axis:

$$(x - h)^2 = 4p(y - k)$$

Focus:

$$(h, k + p)$$

Directrix:

$$y = k - p$$

Horizontal axis:

$$(y - k)^2 = 4p(x - h)$$

Focus:

$$(h + p, k)$$

Directrix:

$$x = h - p$$

Ellipse

Horizontal major axis:

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, \quad a > b$$

Vertical major axis:

$$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1, \quad a > b$$
$$c^2 = a^2 - b^2, \quad e = \frac{c}{a}$$

Hyperbola

Horizontal transverse axis:

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Asymptotes:

$$y - k = \pm \frac{b}{a}(x - h)$$

Vertical transverse axis:

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

Asymptotes:

$$y - k = \pm \frac{a}{b}(x - h)$$
$$c^2 = a^2 + b^2, \quad e = \frac{c}{a}$$

Quick Checks

- Standard form equals 1 for ellipses and hyperbolas.
- The larger denominator under an ellipse gives the major axis direction.
- The positive term in a hyperbola gives the transverse axis direction.
- For parabolas, p controls focus and directrix distance.